THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 4: **Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of neutrons in $^{35}_{17}\mathrm{Cl}$ is: 1

(a) 35 **(b)** 17 **(c)** 18 **(d)** 52

2. Isobars are atoms with the same: 1

(a) atomic number **(b)** number of neutrons

(c) chemical properties **(d)** mass number but different atomic numbers

3. In Rutherford's alpha-particle scattering experiment, most alpha particles passed straight through the gold foil. This shows that: 1

(a) electrons are heavy **(b)** the nucleus is large

(c) the atom is solid **(d)** most of the atom is empty space

4. The number of electrons in $^{27}_{13}\mathrm{Al}$ is: 1

(a) 40 **(b)** 14 **(c)** 13 **(d)** 27

5. Which of the following are isotopes? 1

(a) $^{40}_{18}\mathrm{Ar}$ and $^{40}_{20}\mathrm{Ca}$ **(b)** $^{12}_{6}\mathrm{C}$ and $^{14}_{6}\mathrm{C}$

(c) $^{23}_{11}\mathrm{Na}$ and $^{24}_{12}\mathrm{Mg}$ **(d)** $^{14}_{6}\mathrm{C}$ and $^{14}_{7}\mathrm{N}$

6. Some alpha particles were deflected by large angles, showing that: 1

(a) the foil is thick

(b) atoms are neutral only

(c) the nucleus is positively charged and dense

(d) electrons are positive

7. The maximum number of electrons that can be accommodated in the L shell is: 1

(a) 32 **(b)** 18 **(c)** 8 **(d)** 2

8. Which of the following are isobars? 1

(a) $^{35}_{17}\mathrm{Cl}$ and $^{37}_{17}\mathrm{Cl}$ **(b)** $^{12}_{6}\mathrm{C}$ and $^{13}_{6}\mathrm{C}$

(c) $^{40}_{18}\mathrm{Ar}$ and $^{40}_{20}\mathrm{Ca}$ **(d)** $^{1}_{1}\mathrm{H}$ and $^{2}_{1}\mathrm{H}$

9. Very few alpha particles rebounded. This showed that: 1

(a) the nucleus is very small and dense **(b)** the nucleus is large

(c) there is no nucleus **(d)** the foil was loose

10. The maximum number of electrons in the M shell is: 1

(a) 8 **(b)** 32 **(c)** 2 **(d)** 18

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Write the electronic configuration of magnesium (Z = 12) and its valency. 2

12. State the octet rule. 2

13. State two postulates of Bohr's model. 2

14. What are isotopes? Give an example. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Write the electronic configurations and valencies of: Na, Al, O. 3

16. Explain why isotopes of an element have the same chemical properties. 3

17. Write three properties of canal rays and who discovered them. 3

OR

Write the electronic configuration of the first 10 elements and name the ones with a complete outer shell. 3

18. Explain the rules for filling electrons in the shells with an example of calcium (Z = 20). 3

OR

State three uses of radioactive isotopes. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Write the electronic configuration of the elements with atomic numbers 3, 9, 12, 16, 18 and state valency and the nature (metal or non-metal). 5

OR

Explain the importance of the discovery of the proton, the neutron and the electron, and their charge and mass. 5

20. Explain isotopes with three examples, their identical chemical properties and their uses. 5

OR

Calculate the average atomic mass of an element with two isotopes of masses 10 u (20 per cent) and 11 u (80 per cent). Name the element if Z = 5. 5

END OF QUESTION PAPER — Total: 40 Marks

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